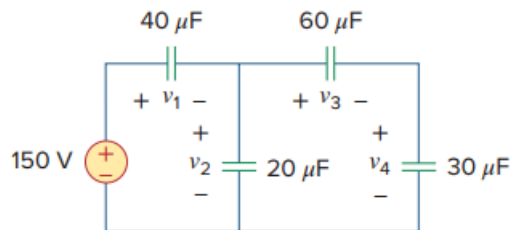


Problem

Find the voltage across each of the capacitors in Fig. 6.20.

Fig. 6.20:



Step-by-step solution

Step 1 of 7

Refer to the circuit in Figure 6.20 in the textbook.

Find the equivalent capacitance with respect to the source, C_{eq} .

In the circuit, $60\ \mu\text{F}$ and $30\ \mu\text{F}$ capacitors are in series and this series combination is in parallel with $20\ \mu\text{F}$ capacitor. So the equivalent capacitance is,

$$\begin{aligned} C_{eq1} &= 20\ \mu\text{F} + \frac{(60\ \mu\text{F})(30\ \mu\text{F})}{60\ \mu\text{F} + 30\ \mu\text{F}} \\ &= 20\ \mu\text{F} + \frac{1800\ \mu\text{F}}{90} \\ &= 20\ \mu\text{F} + 20\ \mu\text{F} \\ &= 40\ \mu\text{F} \end{aligned}$$

Step 2 of 7

The simplified circuit is shown in Figure 1.

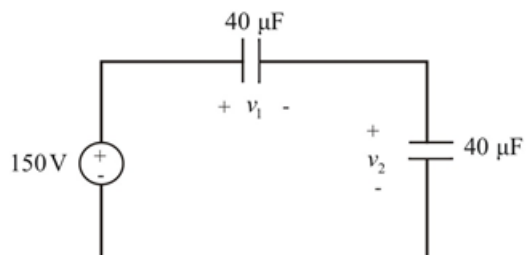


Figure 1

Step 3 of 7

In the circuit shown in Figure 1, two $40\ \mu\text{F}$ capacitances are in series. Hence the equivalent capacitance is,

$$\begin{aligned}C_{eq} &= \frac{(40\ \mu\text{F})(40\ \mu\text{F})}{40\ \mu\text{F} + 40\ \mu\text{F}} \\&= \frac{1600\ \mu\text{F}}{80} \\&= 20\ \mu\text{F}\end{aligned}$$

The expression for the charge is,

$$q = C_{eq}V$$

Substitute $20\ \mu\text{F}$ for C_{eq} and $150\ \text{V}$ for V .

$$\begin{aligned}q &= (20\ \mu\text{F})(150\ \text{V}) \\&= 3\ \text{mC}\end{aligned}$$

Step 4 of 7

Calculate the voltage v_1 .

$$v_1 = \frac{q}{40\ \mu\text{F}}$$

Substitute $3\ \text{mC}$ for q .

$$\begin{aligned}v_1 &= \frac{3\ \text{mC}}{40\ \mu\text{F}} \\&= 75\ \text{V}\end{aligned}$$

Therefore, the voltage v_1 across $40\ \mu\text{F}$ capacitor is $75\ \text{V}$.

Step 5 of 7

Calculate the voltage v_2 .

$$v_2 = \frac{q}{C_{eq1}}$$

Substitute $3\ \text{mC}$ for q $40\ \mu\text{F}$ for C_{eq1} .

$$\begin{aligned}v_2 &= \frac{3\ \text{mC}}{40\ \mu\text{F}} \\&= 75\ \text{V}\end{aligned}$$

Therefore, the voltage v_2 across $20\ \mu\text{F}$ capacitor is $75\ \text{V}$.

Step 6 of 7

Calculate the voltage v_3 by using voltage division rule.

$$\begin{aligned}v_3 &= \left(\frac{30\ \mu\text{F}}{30\ \mu\text{F} + 60\ \mu\text{F}} \right) (75\ \text{V}) \\&= \left(\frac{30\ \mu\text{F}}{90\ \mu\text{F}} \right) (75\ \text{V}) \\&= (0.333)(75\ \text{V}) \\&= 25\ \text{V}\end{aligned}$$

Therefore, the voltage v_3 across $60\ \mu\text{F}$ capacitor is $25\ \text{V}$.

Calculate the voltage v_4 by using voltage division rule.

$$\begin{aligned}v_4 &= \left(\frac{60 \mu\text{F}}{30 \mu\text{F} + 60 \mu\text{F}} \right) (75 \text{ V}) \\&= \left(\frac{60 \mu\text{F}}{90 \mu\text{F}} \right) (75 \text{ V}) \\&= (0.667)(75 \text{ V}) \\&= 50 \text{ V}\end{aligned}$$

Therefore, the voltage v_4 across $30 \mu\text{F}$ capacitor is 50 V.